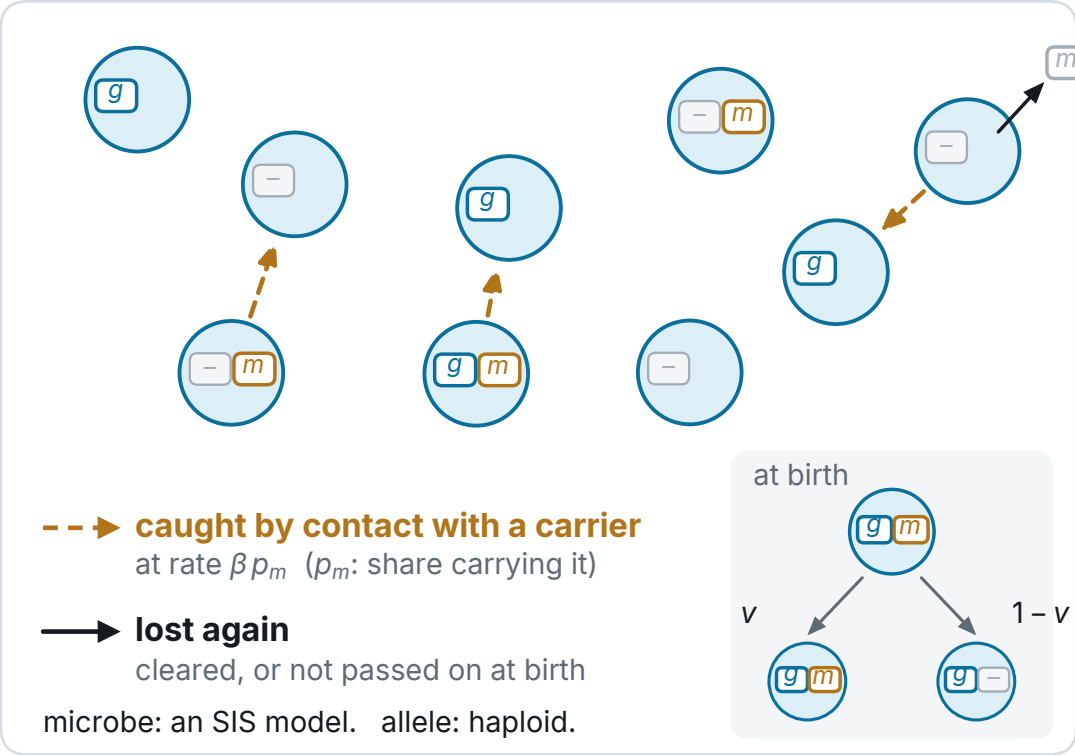
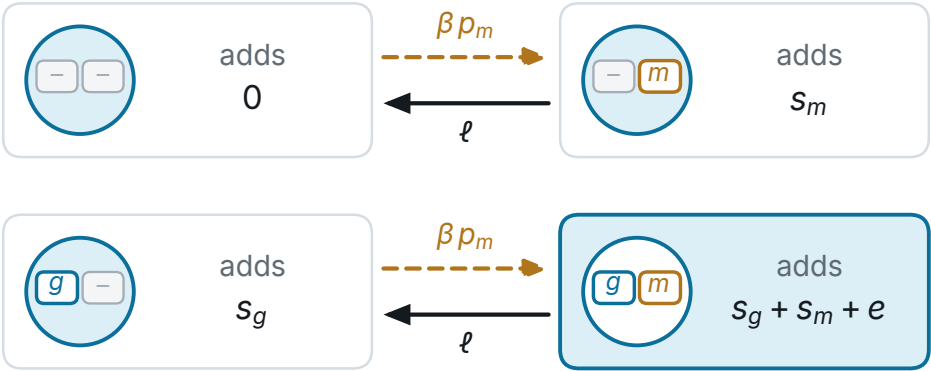


A host allele that is inherited, a microbe that is caught and lost



Four kinds of host



what each kind adds to a growth rate that is otherwise negative

host fitness =
$$\underbrace{s_g}_{\text{the allele}} + \underbrace{s_m}_{\text{direct effect}} + \underbrace{e}_{\text{interaction}}$$

the microbe can help directly, or only together with the allele